

# Illustrative synthetic HFMM environment path

Methodology illustration only: canonical synthetic-environment parameters, not a performance experiment.

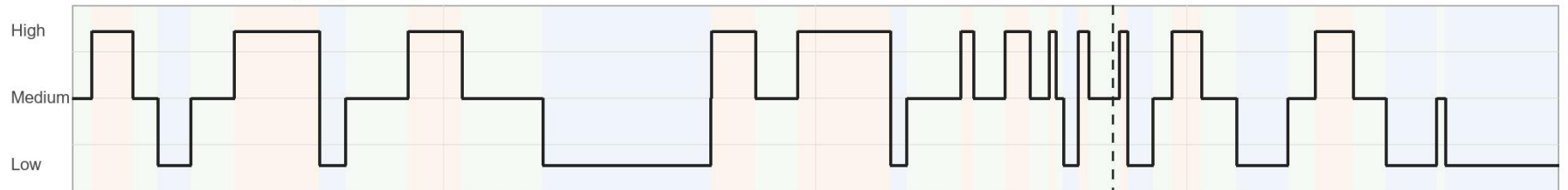
## Synthetic mid-price path



## Rolling realized-volatility signal and true regime volatility



## Latent Markov volatility-regime sequence



Low regime

Medium regime

High regime

Mid /  $\sigma_{\hat{}}$

True volatility

Train-test split

0

2000

4000

6000

8000

Simulation step